ALADIN audit, Phase 5: the single model is built

Written 2026-11-10. Everything is on the `aladin-audit` branch; nothing is on the live site yet.

What was built

File	What it does
scripts/aladin_e.py	Model E: ranks the 500 most-traded stocks by 6-month return, the best 5% form the portfolio, daily NIFTY 500 risk check, paper account with fills at the next open after all costs, degraded state, kill switch, one-day-loss rule, data-error check
data/config/aladin_e.json	Its settings (from the approved spec) and costs
data/config/aladin_e_calibration.json	The measured result of each rank group with 95% ranges (aladin-audit/scripts/e_calibration.py)
tests/test_aladin_e.py	11 tests (below)
.github/workflows/daily-update.yml	Runs model E every evening after the prices are saved
aladin-audit/HOW_ALADIN_WORKS.md	Rewritten for E, with the same three worked examples; the old description kept below it

Every output carries a rank, a measured result with a 95% range, an as-of date and the model version; there is a clear "No signal" state; costs are in every fill; position size is equal weight (about 2.8%), capped at 5% (a quarter of the measured Kelly fraction is 8%); loss limits per trade (the position cap) and per day (no buys after a 4% one-day fall); a kill switch at −25% and a degraded status when data is stale.

Tests (11, all passing)

Test	Phase 2 issue it guards against
Ranks use only data up to the signal day	date alignment / look-ahead
Orders fill at the next open, never at the signal close	date alignment
Running twice on the same evening changes nothing	repeated nightly runs (found in Phase 1)
Stale prices make the model degraded with no orders	missing / stale data
Index below its 200-day average halves the invested share	risk rule

Test	Phase 2 issue it guards against
Kill switch stops buying after −25%	the owner's limit
Costs include STT, stamp, GST, slippage and the DP charge	costs
Position size never above the cap	sizing
Calibration groups ordered, each average inside its range	calibration
No price target in the page's data; version stated	the "target" display issue
A one-day jump beyond any price band is treated as a data error	unadjusted splits / demergers

The Phase 2 display bugs on the old pages (red stop, < 50% = red, calendar shown as a reason) were fixed on main on 2026-11-08 and have their own test (tests/test_aladin_model.py).

Before and after (same dates: Jan 2017 to May 2026, after costs, ₹1,00,000 start)

	Per year	Worst fall	Sharpe	Trades	Won after costs	Net per trade	₹1 lakh became
E as specified (70% invested)	**25.2%**	**−24.7%**	**1.37**	702	46.7%	+9.6%	₹7.85 lakh
E fully invested	33.1%	−36.3%	1.27	702	46.7%	+9.6%	₹13.7 lakh
Old: weekly BUY rule (live until now)	13.2%	−37.1%	0.62	1,746	49.3%	+0.3%	₹3.12 lakh
Old: ALADIN 1, 10-day, top tenth	9.4%	−40.5%	0.61	5,253	51.6%	+0.6%	₹2.28 lakh
NIFTY 500 (price index)	14.0%	−38.3%	0.88	–	–	–	–

What this means: E fits the −25% limit and, in this test, earned far more per trade with far fewer trades. The old weekly rule's result depends heavily on its start date (7.1% a year in Phase 4's walk-forward from 2015, 13.2% here from 2017): with about five stocks a week, luck dominates.

Limits that still apply: companies listed today only (about half of E's tested edge may be survivorship), no result passed the strict luck test, the last six months were flat, and E wins on fewer than half its trades: it relies on a minority of large winners.

How to undo

All on the `aladin-audit` branch; nothing merged. Delete the branch or revert its Phase 5 commit.

Next: Phase 6 (needs your approval)

The public ALADIN tab shows only E (market-state strip, ranked list with "No signal", plain reasons, model health, paper log, alerts). The old models leave every public page and move inside ALADIN BOT with their data encrypted with the same code; they keep running and learning every night. The paper account starts on the first nightly run after merging.